Prerequisite

To read the mind waves, an EEG sensor needs to be connected with the Raspberry Pi. To further process those signals, a Python module is required. So, install NeuroPy library to connect with the EEG sensor and get the measured value. Since the obtained data also has to be streamed over Bluetooth, you would need pyserial library for connecting the EEG data over the Bluetooth serial port.

To move the servo motors, gpiozero library needs to be installed to control the GPIO pins. To install this and above-mentioned Python modules, open the Linux terminal and run following commands:

```
sudo pip3 install NeuroPy
sudo pip3 install pyserial
sudo pip3 instal gpiozero
```

Fig. 4: Robot hand top cover

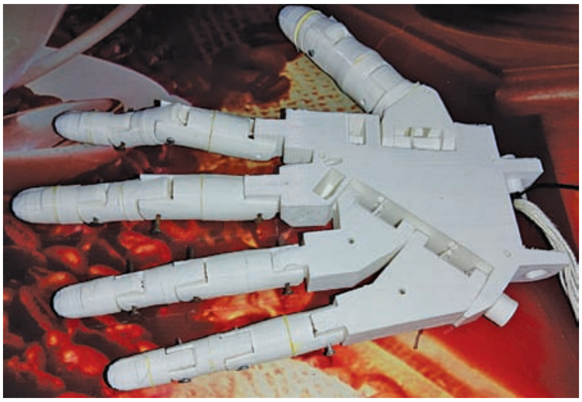

Fig. 5: Robot hand-assembled fingers

```
servoselect=0;
while True:
    #print ("meditation="+str (neuropy.meditation))
    print ("atteb="+str (neuropy.attention))
    if neuropy.attention > 60:
        if servoselect == 1:
            servo1.angle=170
            print ("moving robot arm finger 1")
        if servoselect == 2:
            servo2.angle=170
            print ("moving robot arm finger 2")
        if servoselect == 3:
            servo3.angle=170
            print ("Moving robot wrist")
        if servoselect == 4:
            print ("Moving robot thumb")
            servo4.angle=170
    if neuropy.attention < 60:
        if servoselect == 1:
            servo1.angle=0
            print ("moving robot arm finger 1 position")
        if servoselect == 2:
            servo2.angle=0
            print ("moving robot arm finger 2 position")
        if servoselect == 3:
            servo3.angle=0
            print ("Moving robot wrist position")
        if servoselect == 4:
            print ("Moving robot thumb Position")
            servo4.angle=0


    #print ("rawvalue="+str (neuropy.rawValue))
```

Fig. 7: Code for fingers movement

```
robotarmvraincontrolallfingersver_1.py - /home/pi/N_eropy/robotarmvraincontrolallfingersver_1.py (3.7.3)
File Edit Format Run Options Window Help
from NeuroPy import NeuroPy
from gpiozero import LED
from time import sleep
import time
import os
from gpiozero import LED
from gpiozero import AngularServo
from aiy.pins import PIN_A
from aiy.pins import PIN_B
from aiy.pins import PIN_C
from aiy.pins import PIN_D
from aiy.pins import LED_1
neuropy = NeuroPy()
neuropy.start()

led = LED(LED_1)

servo1 = AngularServo(PIN_A, min_angle=0, max_angle=180, min_pulse_width=.0005,
servo2 = AngularServo(PIN_B, min_angle=0, max_angle=180, min_pulse_width=.0005,
servo3 = AngularServo(PIN_C, min_angle=0, max_angle=180, min_pulse_width=.0005,
servo4 = AngularServo(PIN_D, min_angle=0, max_angle=180, min_pulse_width=.0005,

start_time = 0
blinked = False #if blink is detected
last_blink_time = 0
double_blink = False
triple_blink = False
servoselect=0;
```

Fig. 6: Code importing Python modules

Coding

For coding, first import the NeuroPy, pyserial, gpiozero, and other libraries into the code. Then define the servo min-max and PWM range values along with the GPIO pin number of the Raspberry Pi to control them.

Create variables for storing the servo motor values, which will accordingly control the fingers or joints of the robotic arm using brain waves. Start NeuroPy communication so that your EEG sensor, NeuroSky, connects to Bluetooth and establishes communication with the serial port for data streaming.

In next part of the code, create a while loop to obtain the concentration level on a particular thought that is based on data from brain waves such as Alpha, Beta, Theta,